

Louis Bensard

CTO, ML/AI Lead

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CTO and ML/AI leader with 7+ years building forecasting and decision-intelligence systems from research to production. Proven track record owning the full technical stack — machine learning, data infrastructure, and product engineering — as a startup co-founder and at enterprise scale. Skilled at translating ambiguous business problems into shipped ML/AI solutions and leading technical teams that deliver.

PROFESSIONAL EXPERIENCE

SIGHTLINE OS

New York, NY

Cofounder & CTO (prev. Head of Data & AI)

Aug. 2025 – Current

- **AI/ML Strategy & Forecasting Architecture:** Own AI/ML strategy end-to-end; designed and built a production time series forecasting engine spanning thousands of item-level series, achieving sub-5% wMAPE by building custom PyTorch Transformer (PatchTST) architecture with hurdle modeling for intermittent demand.
- **ML Lifecycle & MLOps:** Built the full ML lifecycle in-house: feature engineering, hyperparameter optimization, multi-seed ensembling, and GPU model training on AWS Batch with Weights & Biases experiment tracking.
- **Decision Intelligence Systems:** Architected a decision-intelligence layer that converts raw model output into ranked, explainable recommendations using composite impact scoring and next-best-action logic.
- **Data Platform & Infrastructure:** Designed and built the data platform powering all ML: multi-source ETL pipelines with data-quality validation and idempotent bulk loading into PostgreSQL, processing millions of rows on AWS (S3, Lambda, SQS).
- **Full-Stack Product Engineering:** Own infrastructure and technical architecture for the customer-facing product in Next.js/TypeScript, multi-tenant — operationalizing ML output for enterprise customers.
- **Technical Leadership:** Own the full technical org as CTO, managing a team of 2 (Sr. Full-Stack Engineer, Sr. Data Engineer); instituted engineering standards (testing, CI/CD, documentation) and defined the data & AI roadmap and technical hiring process.

Tech: Python (PyTorch, scikit-learn, statsmodels), SQL/PostgreSQL, AWS (Batch, Lambda, S3, ECS), Docker, Next.js/TypeScript, MLOps, CI/CD

EKIMETRICS

New York, NY

Manager, Data Science

Jul. 2024 – Aug. 2025

- **Strategic ML Consulting:** Lead cross-functional teams to deliver strategic data science solutions for Fortune 500 clients across sectors like retail and beauty.
- **Predictive Modeling & Deployment:** Manage end-to-end projects, from scoping to deployment, generating \$20M+ in business impact through data-driven strategies.
- **Client Engagement & Delivery:** Serve as primary client contact for 10+ strategic accounts; present insights to C-level stakeholders and align on business objectives.
- **Team & IP Development:** Lead a team of 9+ data scientists across multiple projects; contribute to internal IP/tooling that reduces project ramp-up time by 25% and drives consistency across engagements.

Tech: Python (Scikit-learn, XGBoost, Pandas), SQL, Bayesian Regression, ML Ops, Git, Data Visualization

STRATOS AVIATION

New York, NY

Founder & Lead ML Engineer

Jul. 2023 – Dec. 2024

- **Design & Full-Stack:** Independently designed and built Stratos, a data-driven pilot-centric iOS app in Swift
- **Flight Data Analytics with KNN:** Deployed KNN models on time-series geospatial data processing to provide custom flight recommendations, enhancing pilot decision-making by 75%.
- **Machine Learning Pipelines & Deployment:** Automated data preprocessing, feature selection (Boruta, RFE), and model evaluation pipelines using scikit-learn for integration with Core ML models in production.

Tech: Swift, Python (PyTorch, TensorFlow, Pandas, Scikit), LMM, KNN, NLP, SVM, Neural Nets, Git, GCP, REST APIs, PCA

IPSOS MMA, INC.
Manager & Tech Lead, Data Science

New York, NY
Jul. 2019 – Jun. 2023

- **Bayesian Marketing Mix Modeling:** Applied Bayesian Mix Models using PyMC3 and scikit-learn in Python to build robust Marketing Mix Models (MMM), optimizing Time Series KPI data for Fortune 500 clients.
- **Predictive Analytics at Scale:** Developed and deployed hundreds of predictive models using scikit-learn, XGBoost, and TensorFlow with 10B+ features, integrating Marketing, Macroeconomic, and Business Intel. data.
- **Model Interpretability & Impact:** Delivered strategic insights by leveraging SHAP, LIME, and interpretable ML models for C-suite executives, managing projects worth \$4MM-\$6MM annually.

Tech: Python (PyTorch, Pandas, Scikit, NumPy), Bayesian Marketing Mix Models, Linear & Logistic Regression, SVM, GBM, Git, AWS, REST APIs, MCMC, PCA

PANASONIC AVIATION CORPORATION
Machine Learning Engineer

Fullerton, CA
Jan. 2019 – May 2019

- **System Design & Leadership:** Created production-grade ML algorithms (scikit-learn, TensorFlow) to build an in-flight recommendation engine, optimizing airlines' movie selection using passenger viewing history. Led a 7-person team and optimized in-flight entertainment selection, elevating satisfaction and reducing costs by 45%.
- **Python & Machine Learning:** Predicted passenger viewing preferences with over 85% accuracy using custom-built ML algorithms (GMB) on a 10Mx50 dataset, enabling efficient airline media portfolio strategies.

Tech: Python (TensorFlow, Pandas, Scikit, NumPy, Statsmodels, XGBoost), PCA, KNNs, GBM, Git

EDUCATION

CALIFORNIA STATE UNIVERSITY, FULLERTON
Master of Science, Statistics

Fullerton, CA
Aug. 2017 – May 2019

- summa cum laude, 4.0 GPA, Honor Society: Phi Kappa Phi, Top of class
- Machine Learning & AI, Advanced Theory in Probability and Statistics, Deep Learning, Reinforcement Learning, Algorithms, Multivariate Analysis, Bayesian Analysis

SORBONNE UNIVERSITY
Bachelor of Science, Mathematics

Paris, France
Sept. 2014 – Jun. 2017

- Linear Algebra, Mathematical Probability, Discrete Math, Advanced Calculus, Complex Analysis, Abstract Algebra

SORBONNE UNIVERSITY
Bachelor of Science, Computer Science

Paris, France
Sept. 2014 – Jun. 2017

- Software Engineering, Data Structures & Algorithms, OS, Computer Networks, Theory of Computation

TECHNICAL SKILLS

- **AI/ML (6 years):** Time Series Forecasting, Transformers (PatchTST), Deep learning (CNNs, RNNs), Neural Networks, Gradient Boosting Machines (GBM), Support Vector Machines (SVM), Decision Trees, Clustering, KNNs, Linear & Logistic Regression, NLP, Reinforcement Learning, LMM, MCMC, PCA, Hyperparameter Optimization (GridSearchCV, Bayesian), Feature Engineering, Model Evaluation & Backtesting, MLOps.
- **Python (7 years):** Deep learning frameworks (TensorFlow, PyTorch), NumPy, Pandas, Scikit, Seaborn, Statsmodels, XGBoost, Flask, Docker, GCP, AWS, Agile, Git, CI/CD, Testing.
- **Mathematics (15 years):** Naturally gifted from a young age. Fluent in Linear Algebra, Probability Theory.
- **Marketing Mix Modeling & Multi-Touch Attribution (6 years):** Multivariate regression, time-series modeling, Bayesian statistics, ROI analysis, Forecasting, Optimization, Strategic Results C-suite delivery, MMM problem solving, MMM model fine tuning, data & model validation, data processing, SQL, Excel
- **Technical Leadership (4 years):** Technical Strategy & Architecture (0→1), Full-Stack System Ownership, ML Platform & Roadmap, Engineering Best Practices, Cross-Functional Delivery, Technical Hiring & Team Building.

INTERESTS & LANGUAGES

- **Aviation:** Private Pilot (PPL) since 2022
- **Expanding the mind:** Learning, Traveling, Philosophy
- **Mathematics & Tech:** Latest AI Research, Tech Startup, Rock climbing: Competed in France at National level
- **Polyglot:** English (Fluent), French (Fluent / Native), Spanish (Proficient)